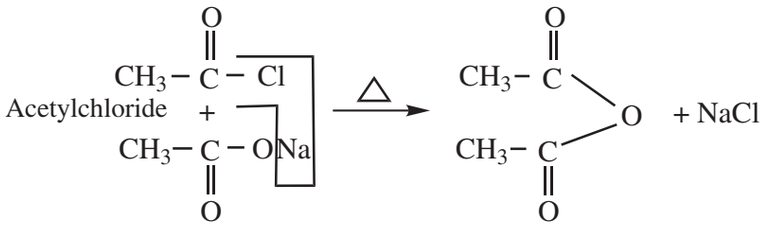
2. By reaction of acid halide with a salt of carboxylic acids.

Acid chlorides on heating with sodium salt of carboxylic acids gives corresponding anhydride.

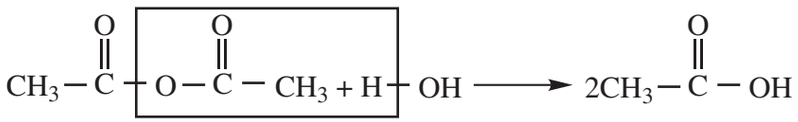


Sodium acetate Acetic anhydride

Chemical properties

1. Hydrolysis

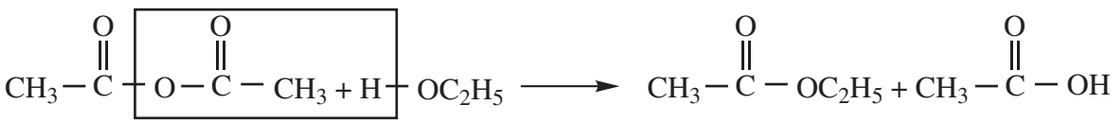
Acid anhydride are slowly hydrolysed, by water to form corresponding carboxylic acids.



Acetic anhydride Acetic acid

2. Reaction with alcohol

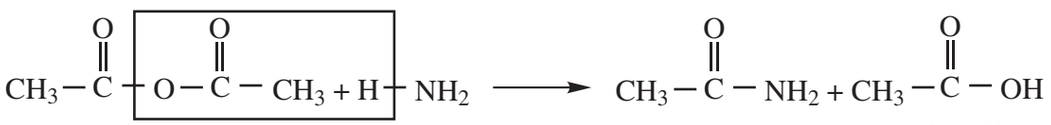
Acid anhydride reacts with alcohols to form esters.



Acetic anhydride Ethylalcohol Ethyl acetate Acetic acid

3. Reaction with ammonia

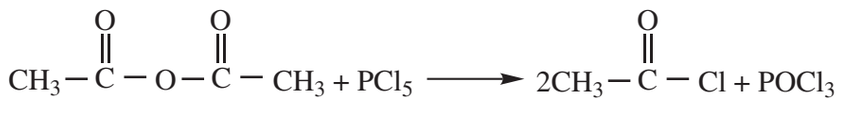
Acid anhydride reacts with ammonia to form amides.



Acetic anhydride Ammonia Acetamide Acetic acid

4. Reaction with PCl_5

Acid anhydride reacts with PCl_5 to form acyl chlorides.



Acetic anhydride Acetyl chloride

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